

PERSONAL PORTFOLIO OF PROJECTS

Project	What I did (1-liner)	Tech stack	Link
Fatal-Crash Data Warehouse	Built Kimball star in PostgreSQL; Python ETL; Tableau dashboards now live with WA road-safety team.	PostgreSQL 13 · pandas/NumPy · Tableau	Github Link
Crash Graph Analytics	Ingested 10 yrs fatal-crash data into Neo4j; Node2Vec + Random Forest flags high-risk LGAs (ROC-AUC 0.88).	Neo4j 5 · APOC/GDS · Python	Github Link
PySpark Loan-Risk (Databricks)	Cleaned 130 k Lending-Club loans; Spark ML pipeline & SQL KPIs drive credit insights.	Spark 3 · MLlib · SQL	Github Link
Distributed XGBoost @ Spark	Colab Spark cluster trains xgboost4j-spark on 580 k Covertypes rows (94 % accuracy).	Spark · xgboost4j · Python	Github Link
Device-Health Predictor	5.6 M telemetry rows → Random-Forest & Keras NN double baseline κ for re-image forecasts.	pandas · scikit-learn · TensorFlow	Github Link
Aspect-Sentiment Transformers	Three custom PyTorch Transformers (Optuna-tuned) hit 65 % test accuracy on restaurant reviews.	PyTorch 2 · Optuna · NLTK	Github Link
AAPL 30-Day LSTM Forecaster	3-layer Keras LSTM predicts next-30-day closes (MAE ≈ \$1.8).	TensorFlow/Keras · pandas-datareader	Github Link
Weighted-Hybrid Movie Recommender	IMDb weight + TF-IDF similarity returns “more-like-this” in < 50 ms.	pandas/NumPy · scikit-learn	Github Link
YouTube Insight Explorer & Earnings Modeller	Dual Shiny apps give self-serve EDA and live model lab for 995 channels + GDP.	R · tidyverse · Shiny/plotly	Youtube Insight Shiny, Code & Earnings Modeller Shiny, Code (Both Video)
WA Economic Power BI Suite	PDFs → Excel → star model; DAX KPIs & AI visuals summarise WA macro-economy.	Power BI Desktop · Power Query · DAX	WA_Dashboard.pbix & Video Explanation
SmartPhone-Rent SQLite DB	Schema with Luhn-checked IMEI; trigger auto-prices rentals, fully indexed.	SQLite · SQL triggers/views	Github Link